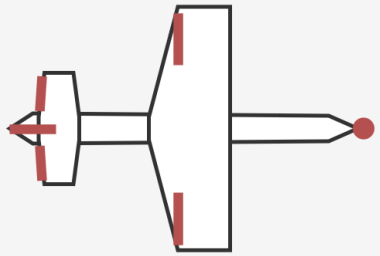
1 ALFA flight logs



ailerons, elevator, rudder (stuck) | engine (off)

Autopilot (Pixhawk) —cmd→ Airframe + sensors
Autopilot (Pixhawk) ←meas— Airframe + sensors

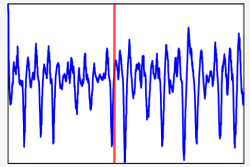
46 real flights, 10 Hz common grid
servo outputs (rc-out) excluded: they reveal the injected fault

2 Physics-informed features

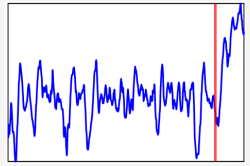
RES r = cmd − meas
roll, pitch, yaw, airspeed, altitude, cross-track, velocity, throttle

CMD commanded inputs
roll, pitch, airspeed commands

EFF control effectiveness
gain and correlation: error vs. body rate



roll residual, aileron fault



airspeed residual, engine fault

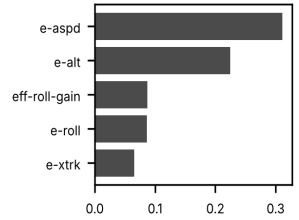
3 s causal windows (0.5 s step)
mean, std, max, slope, last 0.5 s
⇒ 71 features per window

3 Detection, isolation, explanation

Bagged trees (100)
P(class | window)

Alarm: P(fault) ≥ τ
for K windows
Engine | Aileron | Rudder | Elevator

SHAP audit



compare with a priori fault signatures

top-1 hit, expected mass
⇒ consistent or shortcut

Leave-one-flight-out evaluation
detection rate, isolation, delay, false alarms per hour